Andres Flores Nieto

Mechatronics Engineer

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Profile

Dynamic and innovative professional with experience for their leadership abilities and participation in national competitions, demonstrating a commitment to continuous improvement and a passion for developing impactful technological solutions.

Experience History

- Participate in the VEX Robotics Competition

Working in a team, you addressed complex robotic challenges using creative and unconventional design solutions. This experience was key to developing skills in prototyping, technical problem-solving, and critical thinking, which were crucial for creating a unique and functional robot.

- Participate in competitions First (For Inspiration and Recognition of Science and Technology)

Collaborated with a multidisciplinary university team to design, build, and program a robot for a competitive challenge. By applying advanced engineering principles, you optimized the robot's performance to achieve superior results. This experience highlights your agile problem-solving skills and strong commitment to meeting project goals.

- Participate in the Innovation Meetup Competition

Led a university team to solve a real-world business problem. Using a structured approach, you guided the team from ideation to the development of a creative, implementable solution. This demonstrates your strong capacity for identifying core issues and developing innovative strategies that align with business objectives.

Projects

- 6 Degree of Freedom Robotic Arm

A 6 DOF robotic arm was developed starting from a conceptual design evaluated among several alternatives. The selected model was designed in CAD, manufactured by 3D printing, and assembled with the necessary electronic components. The system was programmed in Arduino to control each joint of the arm individually.

- Mobile Robot Controlled by Bluetooth and Wi-Fi

A mobile carriage was developed using an ESP-32 board as the main control system. Electric motors, power electronics, and a wireless communication system were integrated. The robot was programmed to receive commands from an app via Bluetooth and also from a web interface via Wi-Fi. Connectivity, motion control, and system response tests were performed.

- Conveyor Belt Programmed with PLC

A small-scale conveyor belt was built using basic materials and electromechanical integration. The system was programmed using ladder logic to control start, stop, actuator activation, emergency stop, and operational light signaling. The system demonstrated functional operation by transporting and filling a glass using a motorized mechanism.

- Zumo-type Robot

A Zumo-type robot was developed from a CAD design that was then manufactured by 3D printing. The system integrates control electronics, actuators, and sensors for environment perception. The robot's behavior was programmed and an operating strategy was defined for its operation.

- ROS2 Mobile Robot Control

A ROS2-based mobile robot control system was developed and tested in both simulation and a real robot environment. The project included creating Bash scripts and custom robot functions to read sensor data, monitor odometry, and control motion. A naive obstacle avoidance algorithm was implemented to navigate autonomously while maintaining robot statistics during operation

Education

Currently Studying Mechatronics Engineering at TecMilenio University | 6th Semester

Certificates

- Solid Edge Certified Associate Level I
- Certificate in Public Leadership
- Certificate in Additive Manufacturing (in course)
- Linux for Robotics (in course)
- Python for Robotics (in course)

Knowledge

- Foundry & Casting Fundamentals
- Additive Manufacturing
- Advanced Excel
- Python programming
- C++ programming
- Skilled Tinkercad
- Skilled Solid Edge
- Skilled Siemens NX
- Skilled Autodesk Fusión
- Electronic circuit simulation using NI Multisim
- PLC programming
- Programming in LabView
- Fundamental Welding Skills
- Software CAM Development
- CNC Machining
- Turning Processes
- Electronic Circuit Prototyping & Assembly
- Tecnomatix Plant Simulation Fundamentals
- Basic GitHub repository management
- ROS2 Development
- Linux CLI & Bash

Soft Skills

- Teamwork
- Leadership
- Responsibility
- Problem-solving
- Creativity
- Adaptability
- Initiative in different challenges
- Time management

Languages

- Spanish - Native
- English - C1